Mustakim Shikalgar

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Education

Arizona State University — Tempe, AZ
Master of Science in Software Engineering — GPA: 3.77/4.0 — *May 2026*

Technical Skills

Languages: Python, Java, TypeScript/JavaScript, C++, Go, SQL
Backend & Systems: FastAPI, Django, Node.js, gRPC, REST APIs, microservices, distributed systems, object-oriented design, async programming
AI & ML: LangGraph, LangChain, LangSmith, RAG pipelines, LLM fine-tuning, prompt engineering, multi-agent systems, PyTorch, scikit-learn
Infrastructure: Kubernetes, Docker, Prometheus, OpenTelemetry, Redis, NATS JetStream, Grafana, CI/CD
Cloud & Data: Google Cloud Platform, AWS, PostgreSQL, pgvector, MongoDB, Qdrant, Neo4j

Professional Experience

AI Software Engineer — Jan 2026 – May 2026
METY Legal AI Platform, MyEdMaster (Industry Capstone) — *Tempe, AZ*

- Owned end-to-end technical design on the QnA backend, knowledge-profiling system, and production AI architecture for a **6-person cross-functional team** across 7 Agile sprints; demoed to faculty, industry judges, and sponsor stakeholders at ASU SER517 Innovation Showcase
- Reduced per-query LLM cost **82%** ($0.024 → $0.0044/message) by profiling with LangSmith, eliminating an unconditional GPT-4o clarification node, and routing inference to GPT-4o-mini (17× cheaper)
- Designed a **5-node LangGraph pipeline** (RAG → Extraction → Anonymization → Reasoning → Formatter) with strict two-layer separation: stateful Django for session and data ownership, zero-credential stateless FastAPI AI layer for horizontal scale; containerized with Docker and orchestrated via Kubernetes
- Engineered async **FSPR knowledge profiling** across 20 legal topics using background daemon threads, adding **zero user-facing latency**; implemented rolling summarization with tiktoken to prevent context-window overflow

Projects

AegisFlow | *Python, FastAPI, Kubernetes, NATS JetStream, pgvector, Redis* — Feb – May 2026

- Architected a **7-microservice LLM reliability platform** (14 containers, 7 ADRs) that intercepts model responses and routes each through confidence scoring, automated JSON repair, and provider fallback before returning to the caller
- Designed a **4-axis confidence scoring model** (structural parse + Jaccard grounding + LLM critique + rolling provider history) with a diminishing-returns anomaly penalty; drives deterministic ACCEPT / REPAIR / FALLBACK / REJECT routing with **zero application-layer changes**
- Built **sliding-window circuit breaker** (15s–120s exponential cooldown), atomic Redis Lua rate limiting, **11 Prometheus metric families**, OpenTelemetry + Grafana + Tempo distributed tracing, and an event-sourced **replay engine** on NATS JetStream for post-incident root-cause analysis

Distributed Key-Value Store | *Java 17, Raft Consensus, gRPC, Protocol Buffers, Docker* — Jan 2024 – Dec 2025

- Built paper-faithful **Raft consensus** from scratch across a **5-node cluster**: leader election (150–300 ms randomized timeouts), log replication, quorum commit, and tunable consistency (linearizable CP reads vs. AP reads from any replica)
- Benchmarked **17,857 ops/sec** write throughput, **sub-1 ms p99** strong-read latency (10× below target), and **zero data loss** under simultaneous 2-of-5 node failure; full cluster bootstraps and elects a leader within 300 ms

Missing Persons Knowledge Graph | *Python, FastAPI, OWL/RDFLib, SPARQL, React* — Aug – Dec 2024

- Published **"Enhanced Tracking and Reporting of Missing Persons Using Semantic Web Technologies"** at **IEEE COMPSAC 2025** (27% acceptance rate); OWL ontology over **3,559 NamUs records** enables compound SPARQL queries impossible in the original tabular interface
- Replaced managed GraphDB ($50/month) with in-process RDFLib, eliminating infrastructure costs entirely while sustaining **sub-100 ms** SPARQL query latency

Achievements

LeetCode: 881+ problems solved | Knight badge, contest rating 1900+ | Top 3% globally | 130+ Hard | 6+ years
Mathematics Olympiad: Silver Medalist (2015–16)